Types of Research

Research can be classified in many ways, but one practical approach is to group it according to the purpose, method, and nature of data collection. In social sciences, education, and related fields, three major classical types often discussed are survey (descriptive) research, historical/documentary research, and experimental research. These methods differ in what they study, how data are collected, and the kind of conclusions they allow.

1. Survey Research

Survey research is a type of descriptive research that studies present conditions, opinions, attitudes, behaviors, or characteristics of a population. It is commonly used when a researcher wants to know what is happening now rather than what happened in the past or what causes a phenomenon. Survey research often uses questionnaires, interviews, rating scales, or observation to gather data from a sample of people.

Characteristics of survey research

* It focuses on the present situation.
* It describes existing conditions as they are.
* it often uses large samples.
* It may use questionnaires, interviews, or observation.
* It is useful for collecting factual data, opinions, and attitudes.

Types of survey research

1. Questionnaire survey: Data are collected using written questions.
2. Interview survey: Data are collected through oral questioning.
3. Opinion survey: Used to find people's views on an issue.
4. Fact-finding survey: Used to gather facts about a problem or condition.

Merits of survey research

* It covers a wide population.
* It is useful for describing trends and patterns.
* It is relatively economical compared with some other methods.
*It is suitable for educational, social, and political studies.

Limitations of survey research

* It does not explain deep causes very well.

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* Responses may be biased or inaccurate.
* Some people may not return questionnaires.
* It may depend on the honesty and memory of respondents.

Example

A researcher may conduct a survey to find out the study habits of 300-level students in a university. The researcher can ask questions about reading time, preferred study methods, and use of library facilities.

2. Historical/Documentary Research

Historical research is concerned with the past. It involves the collection and critical examination of records, documents, and other sources of evidence in order to reconstruct past events and understand their meaning. Documentary research is closely related because it depends heavily on documents such as books, newspapers, archives, reports, letters, minutes, and official records.

Characteristics of historical/documentary research

* It studies past events.
* It depends on documentary evidence.
* It uses both primary and secondary sources.
* It requires critical judgment about the reliability of sources.
* It seeks to understand how past events influence the present.

Sources of historical data

1. Primary sources: Original records such as letters, diaries, speeches, official documents, photographs, and eyewitness accounts.
2. Secondary sources: Later interpretations such as textbooks, reviews, and scholarly articles.

Steps in historical research

1. Identify the historical problem.
2. Locate relevant sources.
3. Collect documents and records.
4. Critically evaluate the authenticity and reliability of sources.

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5. Interpret the evidence.

6. Draw conclusions and present findings.

Merits of historical/documentary research

* It helps us understand the background of present problems.
* It is useful when direct experimentation is impossible.
* It preserves knowledge from the past.
* It can guide policy and decision-making.

Limitations of historical/documentary research

* Some records may be missing or incomplete.
* Documents may be biased or unreliable.
* The researcher cannot control the events being studied.
* Interpretation may be affected by personal judgment.

Example

A researcher may study the development of education in Nigeria by examining government reports, old school records, and archival documents. This helps explain how present educational policies evolved.

3. Experimental Research

Experimental research is used to determine cause-and-effect relationships. In this type of research, the researcher manipulates one variable and observes its effect on another variable while controlling other factors. It is the best method when a researcher wants to test whether one factor actually causes a change in another.

Characteristics of experimental research

* It involves manipulation of variables.
* It includes control over extraneous factors.
* It tests hypotheses.
* It often uses control and experimental groups.
* It aims at establishing causality.

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Basic variables in experimental research

Independent variable: The variable the researcher changes.

Dependent variable: The variable measured for change.

Control variables: Factors kept constant to avoid interference.

Common steps in experimental research

State the problem.
Formulate a hypothesis.
Identify variables.
Select participants or subjects.
Divide into experimental and control groups.
Apply the treatment or intervention.
Measure the outcome.
Analyze the results and draw conclusions.

Merits of experimental research

1. It helps establish cause-and-effect relationships.
2. It allows control of variables.
3. It is systematic and scientific.
4. It produces strong evidence for testing theories.

Limitations of experimental research

1. It may be expensive and time-consuming.
2. Some experiments are difficult or unethical to carry out on humans.
3. Artificial settings may affect behavior.
4. Findings may not always apply to real-life situations.

Example

A researcher may test whether a new teaching method improves students' performance by comparing one group taught with the new method and another group taught with the traditional method.

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Other Types of Research

Apart from survey, historical/documentary, and experimental research, there are several other important types students should know. These types are often classified based on purpose, approach, or data type.

Descriptive research

Descriptive research aims to describe a phenomenon, situation, or population as it exists. Survey research is often treated as a form of descriptive research. It answers questions such as who, what, where, and when.

Analytical research

Analytical research goes beyond description and tries to examine relationships, explain patterns, or interpret data in greater depth. It often uses existing data and logical analysis to draw meaning.

Correlational research

Correlational research examines the relationship between two or more variables without manipulating them. It shows association, not causation. For example, a study may examine the relationship between study time and exam scores.

Causal-comparative research

This type of research seeks to determine possible causes of existing differences between groups. It is also called ex post facto research because the cause has already occurred. The researcher compares groups after the event has happened.

Qualitative research

Qualitative research explores meanings, experiences, beliefs, and social behavior using non-numerical data. It often uses interviews, focus groups, observation, and document analysis. It is useful for understanding why and how people behave the way they do.

Quantitative research

Quantitative research uses numerical data and statistical analysis. It focuses on measurement, comparison, and testing hypotheses. Survey and experimental research are often quantitative in nature.

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Case study research

Case study research focuses on one person, group, institution, event, or community in detail. It helps the researcher understand a case deeply and holistically. It is common in education, psychology, business, and social science.

Action research

Action research is carried out by practitioners to solve immediate practical problems in their own setting. For example, a teacher may investigate why students are failing a particular topic and then try an intervention.

Ethnographic research

Ethnographic research studies the culture, behavior, and daily life of a group of people over time. It usually involves direct observation and participation in the community.

Comparative research

Comparative research compares two or more groups, institutions, systems, or periods to identify similarities and differences. It is useful in education, politics, and sociology.

Comparison of the Three Main Types

Type	Main focus	Data source	Control of variables	Main purpose
Survey research	Present conditions, opinions, attitudes	Questionnaires, interviews, observation	Low	Describe what is happening now
Historical/documentary research	Past events and records	Documents, archives, records	None over events	Reconstruct and interpret the past
Experimental research	Cause-and-	Tests,	High	Determine

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Type	Main focus	Data source	Control of variables	Main purpose
	effect relationships	measurements, observations		whether one variable causes another

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